

Education

Madison, SD	Dakota State University	Fall 2023 – Spring 2027
• Bachelor of Science in Computer Science and Cyber Operations, GPA: 3.96 • Dean's List, Honors Program, DSU Presidential Champion scholarship, Kappa Sigma Iota Honor Society		

Work Experience

Quantum Security Researcher	Dakota State University	Summer 2024 - Spring 2025
<i>Published paper:</i> Multilevel Post Quantum Encryption for Images with Quantum Fourier Transform (IEEE eIT)		
• Designed a novel multi-level post-quantum image encryption framework (QFT + AES + LWE), securing images against quantum attacks while preserving >40 dB PSNR / 0.95+ SSIM quality while maintaining encryption times under 2.4s for medical, defense, and financial images.		

Quantum AI Researcher	Dakota State University	Fall 2026
<i>Published paper:</i> A Novel Hybrid Quantum-Classical Path Optimization for Methane Detection Using Remote Quantum Intensity Prediction Models (IEEE Access)		
• Co-authored, with Savannah River National Laboratory, a hybrid quantum-classical drone routing system (CatBoost + QAOA) built on SRNL provided RAMS/HYSPLIT dataset, cutting methane leak detection energy use by 50% versus classical baselines. • Implemented a custom Quantum Gaussian Process Regression model (QGPR) in Python, achieving a 20.67% R^2 improvement over classical GPR.		

STEM Academic Tutor	Dakota State University	Fall 2024 – Spring 2025
• Mentored GrowStem scholarship recipients in mathematics and programming fundamentals • Guided students through initial phases of undergraduate research project development		

Technical Experience

Projects
• Shellu: Built a POSIX compatible shell from scratch in Zig, implementing pipes, I/O redirection, background job control, command history, and tab completion alongside a full builtin command set, to gain deeper understanding of shell process management and low level systems programming. • Tracedog: Built a TypeScript program that ingests OpenTelemetry traces and uses LLM narrated causal analysis to autogenerate root cause reports, cutting the manual correlation work engineers do during incident response. • ZNN: A neural network library implemented from scratch in Zig. It includes forward propagation, backpropagation, and gradient descent, to build a first principles understanding of how training actually works under the hood.

Languages and Technologies

• Languages: Zig, C, C++, Python, SQL, JavaScript, TypeScript, Java, C#, HTML, CSS • Technologies: Git, Github, Github Actions, SQLite, PostgreSQL, Svelte, SvelteKit, Docker, PyTorch, Qiskit
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